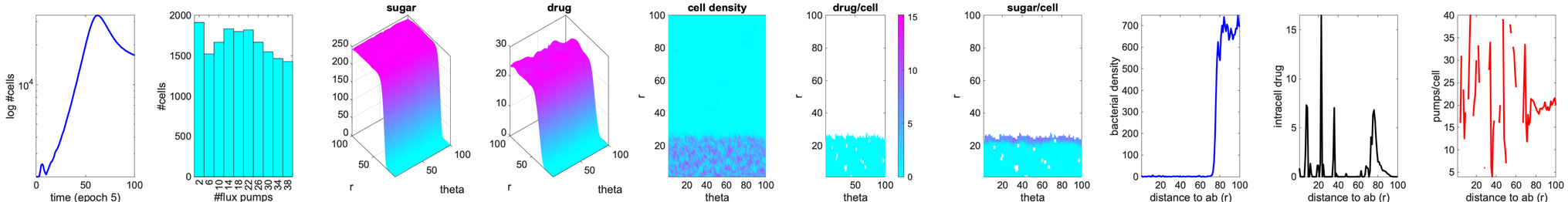
log #cells
10^4
0
50
100
time (epoch 5)
#cells
2000
1500
1000
500
0
2 6 10 14 18 22 26 30 34 38
#flux pumps
sugar
250
200
150
100
50
0
100
50
r
50
100
theta
drug
30
20
10
0
100
50
r
50
100
theta
cell density
100
80
60
40
20
r
20 40 60 80 100
theta
drug/cell
100
80
60
40
20
r
50 100
theta
15
10
5
0
sugar/cell
100
80
60
40
20
r
20 40 60 80 100
theta
bacterial density
700
600
500
400
300
200
100
0
20 40 60 80 100
distance to ab (r)
intracell drug
15
10
5
0
20 40 60 80 100
distance to ab (r)
pumps/cell
40
35
30
25
20
15
10
5
20 40 60 80 100
distance to ab (r)